

Michael Tu

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EDUCATION

Yale University

New Haven, CT

B.S., M.S. in Computer Science

May 2025

- *Relevant coursework:* Algorithms, Linear Algebra, Discrete Mathematics, Intermediate Machine Learning, Zero Knowledge Proofs, Distributed Systems, Operating Systems, Computer Architecture
- *Activities:* Yale Community Kitchen (Coordinator), Yale Men's Water Polo (Business Manager)
- *Awards:* Andy Keidel Undergraduate Endowment Award (2022), Tetelman Fellowship for International Research (2023), Yale Richter Fellowship (2024)

EXPERIENCE

Yale Applied Cryptography Laboratory

New Haven, CT

Research Assistant

Jan 2025 – present

- Researched novel protocol for 2-server PIR based on ORAM which is first to achieve polylog server-side computation with quasi-linear preprocessing
- Led 4 students (2 graduate, 1 undergraduate) to develop custom C++ ring-LWE encryption library, publication pending review

Yale Disaggregated Memory Group

New Haven, CT

Summer Intern - Research Assistant

May 2024 – Dec 2024

- Analyzed memory access distributions for LLama3 and DLRM ml models to construct a performance model for distributed memory architectures, focusing on CXL 3.0 remote memory.
- Discovered significant skew in memory access patterns for models like LLama3, leading ongoing efforts to further characterize these skews across different operational phases.

ICTEAM - Université Catholique de Louvain

Louvain la Neuve, Belgium

Summer Intern – Research Assistant

June 2023 – Aug 2023

- Designed and tested a plugin for the SEMA (Symbolic Execution for Malware Analysis) tool chain to enable symbolic execution of ransomware samples targeting posix compliant systems
- Analyzed GonnaCry and Conti ransoms using manual (Ghidra) and automated techniques (angr)

Yale University Computer Science Department

New Haven, CT

Teaching Assistant (CPSC 323 - Intro to Systems Programming)

Jan 2023 – Present

- Instructed over 200 students about C programming, software design, and operating systems concepts
- Coordinated 17 TAs to host weekly office hours and provide assignment feedback during the semester

PROJECTS

NCCL simulator / Raft Implementation

November 2024

- Implemented Raft consensus protocol in around 600 lines of Golang code
- Designed and implemented NCCL functionality in Golang + used 2-3 simulated GPUs on NCCL to perform distributed computation

Smart Software-L3 Switch

May 2024

- Designed and developed a smart Layer 3 (L3) switch in P4 language, including functionality for ARP table and IP routing table management
- Built a software controller for implementation of modified OSPF protocol to optimize IP packet routing.

CPSC 420 Processor

Jan 2023 - May 2023

- Designed and implemented a 5-stage pipelined processor in Verilog for a custom variant of the MIPS ISA
- Extended processor to support dual instruction fetch, out-of-order commit, and 1-instruction speculation

SKILLS

Language: C++, C, Python, Verilog, Racket, HTML/CSS, Linux/Unix, Java, Assembly, Golang, TCP/IP, Cryptography